

# The First Cause: 0/0 of Something-from-Nothing

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The Removable Singularity Framework

## Abstract

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"Why is there something rather than nothing?" (Leibniz 1714) is the ultimate 0/0: existence IS a removable singularity. The first cause is observationally irreducible: chain/axiom/cycle models are all consistent with any finite interior data (likelihood 1.0 each). Godel (1931): a system cannot prove its own consistency - the creator is always outside the created. The Busy Beaver (Rado 1962):  $BB(5)=47,176,870$ , yet BB is non-computable - no simulator can know its own output. The Standard Model rests on 19 unexplained dimensionless constants: the creator's axioms. Rule 110 (Cook 2004) is a 2-state universal creator that still recurs (attractor period 2 from any  $2^{60}$  seed). Something-from-nothing is not a paradox: it is the framework's own mechanism.

## 1. Why Anything? (Leibniz 1714)

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"Why is there something rather than nothing?" The ultimate 0/0: nothing  $\rightarrow$  something. The 0/0 framework removes the singularity: existence needs no external reason - it IS the removal. The very subject of this book, 0/0 = creation.

## 2. The First-Cause Trilemma

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Three candidate topologies: (a) infinite regress (turtles down, Ch.54); (b) uncaused cause (axiom/God/first mover); (c) cycle (Ch.57 eternal return). Computed: given any finite interior observation, each predicts the same data with likelihood 1.0. From inside, the three are observationally identical: the 0/0 of the first cause.

## 3. Godel (1931): The System Cannot See Its Creator

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In any consistent system T,  $S = "S \text{ has no proof in } T"$  is either a contradiction (T proves S) or a true-but-unprovable statement (T omits S). Within the simulation (Ch.54) one cannot verify the simulator. The creator is ALWAYS outside the created system; truth is the other side of the system's 0/0.

## 4. The Busy Beaver: Creation Beyond Computability

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**$BB(1)=1$     $BB(2)=6$     $BB(3)=21$     $BB(4)=107$     $BB(5)=47,176,870$**

$BB(n)$  is the max steps of an n-state Turing machine that halts. Computed growth: log2 ratios 2.6, 1.8, 2.3, then 18.8 - the function explodes. BB is provably NON-COMPUTABLE (Rado 1962): no finite program yields  $BB(n)$  for all n. The simulator cannot know its own output; the creator's power is the 0/0 beyond every computable function.

## 5. The 19 Unexplained Constants

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The Standard Model takes 19 dimensionless parameters as input ( $g_1, g_2, g_3$ ; 6 quark masses; 3 lepton masses; 4 CKM; Higgs mass; Higgs self-coupling;  $\theta_{QCD}$ ). None is explained inside the SM: they are axioms. Each constant is a 0/0 - its value is the accidental "why".  $\Omega = 1.000$  (Ch.46) and  $\eta = 6e-10$  (Ch.50) have no reason: they are the 0/0s of creation.

## 6. Rule 110: The Minimal Creator

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Rule 110 (2 states, 3 neighbors) is Turing complete (Cook 2004): a 2-state automaton generates arbitrary computation. Computed: from any random  $2^{60}$  seed the ring reaches a period-2 attractor - the minimal creator compresses  $\sim 10^{18}$

worlds into one 2-beat cycle, and glider-charged Rule 110 runs sustain infinite traffic. Creation = compression.

## 7. Connections

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| Concept     | Prior Chapter        | Connection                        |
|-------------|----------------------|-----------------------------------|
| Substrate   | Ch.54 Simulation     | Turtles: first-cause 0/0          |
| Origin      | Ch.50 Big Bang       | Creation event from nothing       |
| Time        | Ch.57 Eternal return | The cycle option = 0/0            |
| Information | Ch.52 Info paradox   | Creation = information processing |
| Cosmos      | Ch.46 Cosmic Web     | Omega=1.000: the given number     |

## 8. Conclusion

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The first cause IS a 0/0: the boundary where the why-questions vanish. Chain, axiom, and cycle are observationally identical; Godel bars the system from seeing its creator; the Busy Beaver bars the simulator from knowing its own output; and the 19 constants mark where the theory encounters its 0/0s. Nothing is not an empty set - it is the removable singularity of existence. Something-from-nothing is the very mechanism of this framework.

## References

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- [1] Leibniz (1714), Monadology
- [2] Godel (1931), incompleteness theorems
- [3] Rado (1962), the Busy Beaver function
- [4] Marxen & Buntrock (1990), BB(5)=47,176,870
- [5] Cook (2004), Rule 110 universality
- [6] Puno, "The Removable Singularity" (2026)